

Doug Hyun Kim

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in LinkedIn

Professional Summary

Ph.D. student focused on hardware-software co-design for efficient and robust deep learning accelerators, with emphasis on Compute-in-Memory (CiM), memory design, and cross-layer modeling of hardware non-idealities. Research spans device, circuit, architecture, and DNN workload levels, combining PyTorch-based algorithm evaluation with SPICE and Cadence-based circuit validation for CMOS and emerging-memory technologies.

Education

Purdue University

Ph.D. in Electrical and Computer Engineering

Aug 2024 – Present

West Lafayette, IN

Hongik University

B.S. in Electronic and Electrical Engineering

Mar 2017 – Feb 2024

Seoul, South Korea

Experience

SK hynix, Future Technology Research Institute

Global Intern, Future Memory Research Lab

Jun 2026 – Aug 2026

Icheon, South Korea

- Developed a cross-layer, device-to-system pathfinding framework for evaluating SRAM, eDRAM/DRAM, and emerging-memory analog Compute-in-Memory (ACIM) designs, connecting workload-level accuracy requirements to hardware evaluation and circuit/device parameter improvement targets.
- Translated DNN workload-level accuracy requirements into circuit and memory-device parameter targets for an emerging-memory ACIM design, mapping the allowable computation-error budget to signal range, device variation, and readout-noise constraints.

Integrated Circuits and Devices Lab

Graduate Research Assistant

Aug 2024 – Present

Advisor: Dr. Sumeet Kumar Gupta

- Built a PyTorch-based cross-layer simulation framework for charge-based Compute-in-Memory (CiM), enabling rapid evaluation of circuit non-idealities, noise, variation, and voltage-scaling effects on DNN inference accuracy with $>10000\times$ speedup over SPICE and $>99\%$ accuracy against circuit-level baselines.
- Proposed and validated a mitigation technique for charge-based CiM that restores near-software accuracy for ViT and ResNet-50 on CIFAR-100 under non-ideal charge-domain accumulation, improving noise tolerance and supporting lower-voltage operation.

ASP-DAC 2026: 31st Asia and South Pacific Design Automation Conference

- Developed and presented a fault-tolerant technique for ternary Compute-in-Memory (TCiM) accelerators at the CoCoSys Annual Review, a Joint University Microelectronics Program (JUMP) 2.0 academic research center co-sponsored by the Semiconductor Research Corporation (SRC) and the Defense Advanced Research Projects Agency (DARPA).

CoCoSys Annual Review 2025: Center for the Co-Design of Cognitive Systems, Atlanta, GA

Publications

- Doug Hyun Kim**, Akul Malhotra, and Sumeet Kumar Gupta, “SCION: A Comprehensive Simulation Framework for Charge-based In-Memory Computing for Rapid Evaluation of Hardware Non-Idealities and DNN Accuracy,” *2026 31st Asia and South Pacific Design Automation Conference (ASP-DAC)*, pp. 829–836, 2026. [DOI](#) [↗](#).

Coursework and Technical Skills

- Coursework:** MOS VLSI Design, AI Hardware, Advanced VLSI Design, Digital Systems Design Automation, Analog Circuit Design, Semiconductor Physics, Semiconductor Memory Technologies, Digital Logic Circuit, Digital Signal Processing
- Design Tools:** Cadence Virtuoso, Layout XL, Spectre, Pegasus, Synopsys HSPICE, Siemens Calibre, COMSOL, Microchip Studio
- Programming Languages:** Python, PyTorch, C, C++, MATLAB, Verilog, SystemVerilog, LaTeX

Projects

2T–2MTJ MRAM Digital Compute-in-Memory Macro | GF 22FDX, Cadence Virtuoso, HSPICE, Layout XL

Apr 2026

- Designed and evaluated a compact 2T–2MTJ MRAM digital Compute-in-Memory (DCIM) bitcell and 64×64 macro for binary matrix-vector multiplication, including write-path verification, row-dependent parasitic analysis, and global variation simulation.

- Reduced projected bitcell area to $0.71 \mu\text{m}^2$, approximately 32% smaller than the 1T-1MTJ reference layout, while estimating one-cycle 64×64 MVM latency of 1 ns at 1 GHz and 171.7 TOPS/W energy efficiency, equivalent to approximately 5.8 fJ/op and 23.9 pJ per MVM.

8+T vs 10T Differential SRAM Evaluation for CiM | *Synopsys HSPICE, 45nm PTM*

Dec 2025

- Constructed a 128×128 distributed-RC SRAM array simulation framework in HSPICE using 45nm CMOS to quantify source-degeneration bottlenecks in 8+T SRAM architectures, identifying 406 mV virtual-ground bounce as the dominant sense-margin limiter.
- Validated a 10T differential bitcell topology for robust in-memory Boolean logic, achieving $> 2\times$ improvement in sense margin and 48% lower dynamic energy (1.13 pJ/op) by eliminating virtual-ground instability.

Compute-in-Memory (CiM) Architecture Full-stack Analysis | *CiMLoop*

May 2025

- Performed full-stack device-to-architecture analysis of CiM accelerators using CiMLoop, evaluating DNN workload mapping, memory hierarchy, dataflow strategies, and circuit-level energy/latency contributors across ResNet18 and GPT-2 workloads.
- Optimized CiM system configurations by analyzing latency, energy, and area trade-offs, identifying workload-dependent bottlenecks across compute arrays, peripheral circuits, and data-movement paths.

Wallace Tree Multiplier and MAC unit | *Cadence Virtuoso, Spectre, Layout XL*

Dec 2024

- Designed and implemented an 8-bit Wallace tree multiplier using NAND-based adders and a transmission-gate XOR structure, completing a full DRC/LVS-clean layout and post-layout evaluation with 1.43 ns worst-case propagation delay and 506.08 fJ energy consumption.
- Integrated the multiplier with a 16-bit ripple-carry adder to build a Multiply-Accumulate (MAC) unit, demonstrating functional scalability from arithmetic datapath primitives to larger accelerator building blocks.

Boolean Satisfiability (SAT) Solver | *Python*

May 2025

- Implemented a complete SAT solver in Python based on the Davis-Putnam-Logemann-Loveland (DPLL) algorithm for digital design automation and logic-reasoning benchmarks, including Random-3-SAT, graph coloring, and SAT-encoded planning.
- Added two-watched literals and Variable State Independent Decaying Sum (VSIDS) decision heuristics, achieving up to $330\times$ speedup and $2.8\times$ memory reduction on complex benchmarks.

Honors

Academic Excellence Scholarship, Hongik University

Sep 2023, Mar 2022

Undergraduate Research Assistant Scholarship, Hongik University

Sep 2022